

Practice Questions for Formulating and Solving Linear Programs

Generated by Scribe.AI

December 31, 2024

1 Questions

1. A furniture manufacturer produces chairs and tables. Each chair requires 2 hours of carpentry and 1 hour of finishing, while each table requires 3 hours of carpentry and 2 hours of finishing. The manufacturer has 100 hours of carpentry time and 60 hours of finishing time available per week. The profit from each chair is 30 and from each table is 40. Formulate this problem as a linear program to maximize profit. What is the objective function?
 - a. Minimize $Z = 2x_c + 3x_t$
 - b. Maximize $Z = 30x_c + 40x_t$
 - c. Maximize $Z = 100x_c + 60x_t$
 - d. Minimize $Z = x_c + x_t$
 - e. Maximize $Z = 2x_c + x_t + 3x_c + 2x_t$
2. Continuing the furniture manufacturer problem, what constraint represents the limitation on the available carpentry hours?
 - a. $x_c + x_t \leq 100$
 - b. $x_c + 2x_t \leq 60$
 - c. $2x_c + 3x_t \leq 100$
 - d. $x_c + 3x_t \leq 100$
 - e. $2x_c + x_t \leq 60$
3. In a linear programming problem, what is the significance of the feasible region, and how does it relate to the optimal solution?
 - a. The feasible region is irrelevant to finding the optimal solution.
 - b. The feasible region contains all points that satisfy the constraints, and the optimal solution is a point within this region that optimizes the objective function.
 - c. The feasible region is only important if the problem is unbounded.
 - d. The feasible region is the set of all possible objective function values.
 - e. The feasible region is only relevant in graphical methods, not in the simplex method.
4. What is the fundamental difference between a linear programming problem and a non-linear programming problem?

- a. Linear programming problems always have a unique solution, while non-linear problems may have multiple solutions.
 - b. Linear programming problems involve only linear functions in the objective function and constraints, while non-linear problems involve at least one non-linear function.
 - c. Linear programming problems are always easier to solve than non-linear problems.
 - d. Linear programming problems always have a feasible region, while non-linear problems may not.
 - e. Linear programming problems only deal with continuous variables, while non-linear problems can handle discrete variables.
5. A company manufactures two products, A and B. Product A requires 2 hours of machine time and 1 hour of labor, while product B requires 1 hour of machine time and 3 hours of labor. The company has 100 hours of machine time and 150 hours of labor available. The profit from each unit of product A is 10, and the profit from each unit of product B is 15. The company wants to determine the optimal production quantities of A and B to maximize profit. Formulate this problem as a linear program. Which of the following correctly represents the objective function and constraints?
- a. Maximize $Z = 10x_A + 15x_B$ subject to $2x_A + x_B \leq 100$, $x_A + 3x_B \leq 150$, $x_A, x_B \geq 0$
 - b. Maximize $Z = 10x_A + 15x_B$ subject to $2x_A + x_B \geq 100$, $x_A + 3x_B \geq 150$, $x_A, x_B \geq 0$
 - c. Maximize $Z = 10x_A + 15x_B$ subject to $2x_A + x_B \leq 100$, $x_A + 3x_B \leq 150$, $x_A, x_B \leq 0$
 - d. Maximize $Z = 10x_A + 15x_B$ subject to $2x_A + x_B = 100$, $x_A + 3x_B = 150$, $x_A, x_B \geq 0$
 - e. Minimize $Z = 10x_A + 15x_B$ subject to $2x_A + x_B \leq 100$, $x_A + 3x_B \leq 150$, $x_A, x_B \geq 0$
6. A company produces two products, X and Y. Product X requires 3 units of raw material A and 2 units of raw material B. Product Y requires 1 unit of raw material A and 4 units of raw material B. The company has 100 units of raw material A and 120 units of raw material B available. Let x and y represent the number of units of products X and Y produced, respectively. What constraint represents the limitation on the availability of raw material A?
- a. $3x + y \leq 100$
 - b. $x + 4y \leq 120$
 - c. $2x + 4y \leq 120$
 - d. $3x + 1y \geq 100$
 - e. $3x + y \geq 100$
7. A farmer has 100 acres of land to plant corn and soybeans. Corn requires 2 hours of labor per acre, and soybeans require 1 hour of labor per acre. The farmer has 150 hours of labor available. Let c and s represent the number of acres of corn and soybeans planted, respectively. What constraint represents the limitation on the total acreage?
- a. $2c + s \leq 150$
 - b. $c + s \leq 100$
 - c. $c + 2s \leq 100$
 - d. $2c + s \geq 150$

- e. $c + s \geq 100$
8. A bakery makes cakes and cookies. Each cake requires 2 cups of flour and 1 egg, while each cookie requires 0.5 cups of flour and 0.25 eggs. The bakery has 50 cups of flour and 20 eggs available. Let c and k represent the number of cakes and cookies made, respectively. What is the constraint representing the limitation on the number of eggs?
- a. $2c + 0.5k \leq 50$
- b. $c + 0.25k \leq 20$
- c. $2c + k \leq 50$
- d. $c + 0.25k \geq 20$
- e. $2c + 0.5k \geq 50$
9. Maximize $Z = 5x_1 + 3x_2$ subject to $x_1 + x_2 \leq 4$, $2x_1 + x_2 \leq 6$, $x_1, x_2 \geq 0$. After introducing slack variables s_1 and s_2 , what is the initial simplex tableau?
- a. None of the above
10. A company produces two products, X and Y. Product X requires 2 units of raw material A and 3 units of raw material B. Product Y requires 4 units of raw material A and 1 unit of raw material B. The company has 100 units of raw material A and 80 units of raw material B available. The profit from each unit of product X is 5, and the profit from each unit of product Y is 7. Let x and y represent the number of units of products X and Y produced, respectively. Formulate this as a linear program to maximize profit. What is the constraint representing the limitation on the availability of raw material B?
- a. $2x + 4y \leq 100$
- b. $3x + y \leq 80$
- c. $2x + y \leq 80$
- d. $3x + 4y \leq 80$
- e. $5x + 7y \leq 180$

2 Answers

1. A furniture manufacturer produces chairs and tables. Each chair requires 2 hours of carpentry and 1 hour of finishing, while each table requires 3 hours of carpentry and 2 hours of finishing. The manufacturer has 100 hours of carpentry time and 60 hours of finishing time available per week. The profit from each chair is 30 and from each table is 40. Formulate this problem as a linear program to maximize profit. What is the objective function?
 - a. Answer A is incorrect. This minimizes a combination of carpentry and table production hours, not profit.
 - b. Answer B is correct. The objective is to maximize profit, and the profit from chairs (x_c) is 30 per chair and from tables (x_t) is 40 per table. Therefore, the objective function is to maximize $Z = 30x_c + 40x_t$.
 - c. Answer C is incorrect. This uses the available hours as the objective, not the profit.
 - d. Answer D is incorrect. This minimizes the total number of chairs and tables produced, not the profit.
 - e. Answer E is incorrect. This incorrectly combines carpentry and finishing hours in the objective function.
2. Continuing the furniture manufacturer problem, what constraint represents the limitation on the available carpentry hours?
 - a. Answer A is incorrect. This constraint does not correctly account for the different carpentry requirements of chairs and tables.
 - b. Answer B is incorrect. This constraint represents the finishing time limitation, not the carpentry time limitation.
 - c. Answer C is correct. Each chair requires 2 hours of carpentry ($2x_c$) and each table requires 3 hours ($3x_t$). The total carpentry hours used must be less than or equal to the available 100 hours. Therefore, the constraint is $2x_c + 3x_t \leq 100$.
 - d. Answer D is incorrect. This constraint incorrectly uses the carpentry hours for tables but not chairs.
 - e. Answer E is incorrect. This constraint incorrectly uses the finishing hours for chairs and carpentry hours for tables.
3. In a linear programming problem, what is the significance of the feasible region, and how does it relate to the optimal solution?
 - a. Answer A is incorrect. The feasible region is fundamental to linear programming; the optimal solution must be feasible.
 - b. Answer B is correct. The feasible region is defined by the constraints of the problem. The optimal solution must lie within this region, representing the best possible outcome given the limitations.
 - c. Answer C is incorrect. The feasible region is crucial regardless of whether the problem is bounded or unbounded.
 - d. Answer D is incorrect. The feasible region is the set of points satisfying the constraints, not the set of objective function values.
 - e. Answer E is incorrect. The feasible region is central to both graphical and algebraic methods for solving linear programs.

4. What is the fundamental difference between a linear programming problem and a non-linear programming problem?
- a. Answer A is incorrect. Both linear and non-linear problems can have unique or multiple solutions.
 - b. Answer B is correct. The defining characteristic is the linearity of the objective function and constraints. Non-linear problems have at least one non-linear term, making them significantly more complex to solve.
 - c. Answer C is incorrect. Non-linear problems are generally much harder to solve than linear problems.
 - d. Answer D is incorrect. Both types of problems can have feasible or infeasible regions.
 - e. Answer E is incorrect. Both linear and non-linear programming can handle continuous and discrete variables (though integer programming is a separate subfield).
5. A company manufactures two products, A and B. Product A requires 2 hours of machine time and 1 hour of labor, while product B requires 1 hour of machine time and 3 hours of labor. The company has 100 hours of machine time and 150 hours of labor available. The profit from each unit of product A is 10, and the profit from each unit of product B is 15. The company wants to determine the optimal production quantities of A and B to maximize profit. Formulate this problem as a linear program. Which of the following correctly represents the objective function and constraints?
- a. The objective function correctly represents maximizing profit. The constraints correctly reflect the limitations on machine time and labor, with the inequalities ensuring that the resource usage does not exceed the available amounts. The non-negativity constraints are also essential.
 - b. The constraints are incorrectly formulated as greater than or equal to inequalities. The available resources cannot exceed the usage.
 - c. The non-negativity constraints are incorrectly formulated as less than or equal to zero. Production quantities cannot be negative.
 - d. The constraints are incorrectly formulated as equalities. It is not necessary to use all available resources.
 - e. The objective function is incorrectly formulated as a minimization problem. The goal is to maximize profit.
6. A company produces two products, X and Y. Product X requires 3 units of raw material A and 2 units of raw material B. Product Y requires 1 unit of raw material A and 4 units of raw material B. The company has 100 units of raw material A and 120 units of raw material B available. Let x and y represent the number of units of products X and Y produced, respectively. What constraint represents the limitation on the availability of raw material A?
- a. The constraint on raw material A is $3x + y \leq 100$ because each unit of product X uses 3 units of A, and each unit of product Y uses 1 unit of A. The total usage cannot exceed the available 100 units.
 - b. This constraint represents the limitation on raw material B, not A.
 - c. This constraint is incorrect; it uses the amount of raw material B for both products X and Y.
 - d. This is an incorrect inequality; it should be less than or equal to, not greater than or equal to.
 - e. This inequality should be less than or equal to, not greater than or equal to.

7. A farmer has 100 acres of land to plant corn and soybeans. Corn requires 2 hours of labor per acre, and soybeans require 1 hour of labor per acre. The farmer has 150 hours of labor available. Let c and s represent the number of acres of corn and soybeans planted, respectively. What constraint represents the limitation on the total acreage?
- This constraint represents the limitation on labor hours, not acreage.
 - The total acreage planted cannot exceed 100 acres, so the constraint is $c + s \leq 100$.
 - This constraint is incorrect; it uses the labor hours for soybeans incorrectly.
 - This inequality should be less than or equal to, not greater than or equal to.
 - This inequality should be less than or equal to, not greater than or equal to.
8. A bakery makes cakes and cookies. Each cake requires 2 cups of flour and 1 egg, while each cookie requires 0.5 cups of flour and 0.25 eggs. The bakery has 50 cups of flour and 20 eggs available. Let c and k represent the number of cakes and cookies made, respectively. What is the constraint representing the limitation on the number of eggs?
- This constraint represents the limitation on flour, not eggs.
 - Each cake uses 1 egg, and each cookie uses 0.25 eggs. The total number of eggs used cannot exceed 20, so the constraint is $c + 0.25k \leq 20$.
 - This constraint is incorrect; it uses the amount of flour for cookies incorrectly.
 - This inequality should be less than or equal to, not greater than or equal to.
 - This inequality should be less than or equal to, not greater than or equal to.
9. Maximize $Z = 5x_1 + 3x_2$ subject to $x_1 + x_2 \leq 4$, $2x_1 + x_2 \leq 6$, $x_1, x_2 \geq 0$. After introducing slack variables s_1 and s_2 , what is the initial simplex tableau?
-
10. A company produces two products, X and Y. Product X requires 2 units of raw material A and 3 units of raw material B. Product Y requires 4 units of raw material A and 1 unit of raw material B. The company has 100 units of raw material A and 80 units of raw material B available. The profit from each unit of product X is 5, and the profit from each unit of product Y is 7. Let x and y represent the number of units of products X and Y produced, respectively. Formulate this as a linear program to maximize profit. What is the constraint representing the limitation on the availability of raw material B?
- This constraint represents the limitation on raw material A, not B.
 - The constraint on raw material B is given by the total usage of B in producing X and Y, which must be less than or equal to the available 80 units. Product X uses 3 units of B per unit produced, and product Y uses 1 unit of B per unit produced. Therefore, the constraint is $3x + y \leq 80$.
 - This constraint incorrectly uses the raw material A coefficients for product X.
 - This constraint incorrectly uses the raw material A coefficients for product Y.
 - This is the objective function, not a constraint.